

# Coexistence of evolutionary strategies

Sang Woo Park<sup>1,\*</sup>

<sup>1</sup> School of Biological Sciences, Seoul National University, Seoul, Korea

\*Corresponding author: sangwoopark@snu.ac.kr

## Abstract

## Introduction

Evolutionary game theory (EGT) provides a powerful framework for predicting which competing strategies are evolutionarily stable. Because the success of each strategy depends on the frequency of other strategies, even simple two-strategy games can yield heterogeneous outcomes, including the dominance of one strategy (prisoner's dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). These differences are typically understood through payoff inequalities, invasion criteria, or stability analysis, but Tarnita and Traulsen (2025) recently argued that an ecological perspective on EGT may yield new, ecological insights. In particular, recasting game dynamics as a coexistence problem immediately raises two important questions: when two strategies coexist, what mechanism stabilizes it? And when coexistence fails, is it because stabilizing effects are absent, or because fitness differences are too large?

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for answering these questions. In particular, MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy's niche can be interpreted as strategic environments generated by the composition of competing strategies and their interactions. Thus, stabilizing niche differences arise when strategies are favored by different strategic environments, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition,

to other competitive systems, allowing one to directly quantify niche and fitness differences of competing entities. Inspired by this work, we present a new theoretical framework for analyzing the coexistence and exclusion of competing strategies by quantifying their niche and fitness differences.

## Mathematical theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where  $x_i$  represents the frequency of strategy  $i$ ,  $\pi_i(\mathbf{x})$  represents the payoff of strategy  $i$  for a given strategy distribution  $\mathbf{x}$ , and  $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$  represents the average payoff. Since strategy frequencies must sum to 1 (i.e.,  $\sum_i x_i = 1$ ), a single-strategy equilibrium dominated by strategy  $i$  can be described by a unit vector  $\mathbf{e}_i$ , where all elements are equal to zero except for the  $i$ -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies  $i$  and  $j$ . However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

For convenience, we refer to  $W_i(\mathbf{x})$  as the multiplicative payoff of strategy  $i$ .

First, we define the niche overlap  $\rho$  between strategies  $i$  and  $j$  as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common,  $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$ , and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor,  $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$ . The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When  $1 - \rho > 0$ , each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition.

Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

67 In other words, strategy  $j$  has an average competitive advantage over strategy  $i$   
68 ( $\kappa_j/\kappa_i > 1$ ) when it performs better, on average, across the environments generated  
69 by itself and its competitor. Then, analogous to MCT, mutual invasion requires the  
70 fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

71 This decomposition allows us to understand how different mechanisms contribute to  
72 the coexistence or exclusion of competing strategies.

## 73 **Two strategy games**

74 As an illustrative example, we begin with classic two strategy games, which can be  
75 described by the following payoff matrix:

$$\begin{pmatrix} P & Q \\ R & S \end{pmatrix} \quad (6)$$

## 76 **Supplementary Text**

## 77 **References**

- 78 Tarnita, C. E. and A. Traulsen (2025). Reconciling ecology and evolutionary game  
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